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C4 — Material Choices

OCR GCSE Chemistry B (Twenty First Century Science, J258) — Chapter C4 · All papers (J258/01–04)

Content marked **(HT)** is tested only on the Higher tier papers (J258/03 and J258/04). Everything else can be tested on both tiers.

C4.1 How Is Data Used to Choose a Material for a Particular Use?

Chemists assess materials by **measuring their physical properties**, then use the data to compare materials and match one to the **specification** of a product (the list of properties it needs). The huge range of materials developed by chemists improves our quality of life **(IaS4)**.

Key Properties

Property	Meaning
Melting point / softening temperature	Temperature at which a solid melts. Polymers (plastics) usually *soften* over a range of temperatures rather than melting sharply.
Electrical conductivity	How well it conducts electricity (conductor or insulator).
Strength in tension	Force needed to break it by pulling (stretching).
Strength in compression	Force needed to break it by squashing.
Stiffness / flexibility	Stiff materials resist bending or stretching; flexible ones bend easily without breaking.
Brittleness	A brittle material breaks suddenly (shatters) without bending first. The opposite is *tough*.

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Hardness	Resistance to being scratched or dented. Test: which material scratches which.
Density	Mass per unit volume (g/cm^3). Low density matters for anything that must be light.
Ease of reshaping	How easily it can be moulded, bent or pressed into a new shape.

Comparing the Main Classes of Material

Material class	Typical properties	Examples and uses
Metals	Conduct electricity and heat; strong in tension; malleable and ductile; high melting point; usually dense	Steel girders, copper wire, aluminium cans
Clay ceramics	Hard; strong in compression but weak in tension; brittle; very high melting point; insulators	Bricks, tiles, pottery, sinks
Glass	Transparent; hard; brittle; strong in compression; insulator; softens when very hot so can be shaped	Windows, bottles
Polymers	Low density; flexible (many); insulators; low softening temperatures; easy to reshape; often weaker than metals	Bags, bottles, pipes, cable insulation
Composites	Combine the properties of two or more materials: a *reinforcement* held in a *matrix*	Reinforced concrete, fibreglass, carbon-fibre-reinforced plastic

The data below are typical approximate values (real values depend on the grade of material) for comparing materials quantitatively:

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Material	Density / g/cm³	Melting or softening point / °C	Tensile strength / MPa	Conducts electricity?
Mild steel	7.8	~1450 (melts)	~400	Yes
Aluminium alloy	2.7	~600 (melts)	~300	Yes
Glass	2.5	~700 (softens)	~50	No
Clay ceramic (brick)	~2.0	>1500	~5 (much stronger in compression)	No
Poly(ethene), low density	0.92	~100 (softens)	~10	No
Carbon-fibre composite	1.6	— (resin softens ~150)	~600 along fibres	Partly
Concrete	2.4	—	~3 (about 30–40 in compression)	No

Worked Example: A bicycle frame must be strong, stiff and light. Use the table to choose between steel and carbon-fibre composite.
Strength per unit density: steel 400 ÷ 7.8 ≈ 51; carbon fibre 600 ÷ 1.6 ≈ 375.
Carbon-fibre composite gives about **7 times** the strength for the same mass, so it is better for a racing bike — though steel is cheaper and easier to repair, so it suits everyday bikes.

Composites

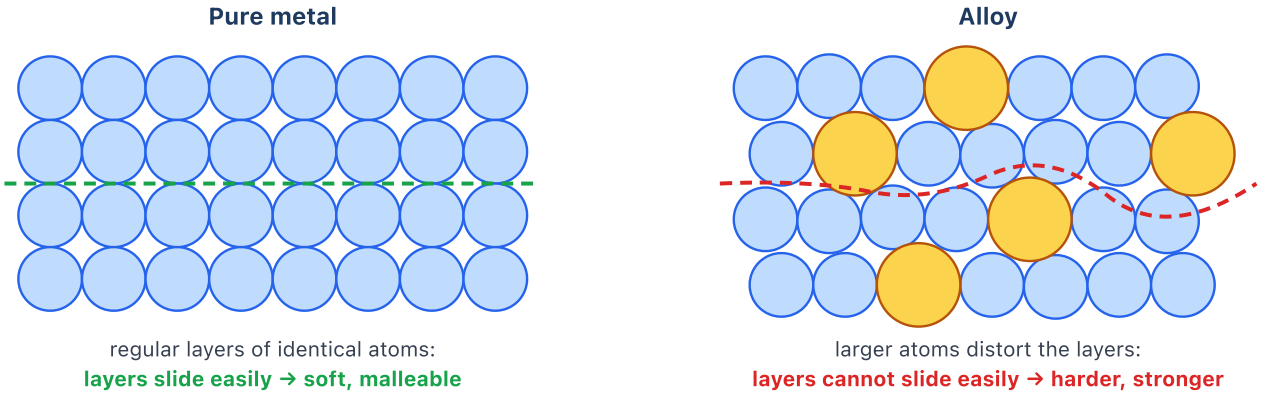
A **composite** combines two or more materials to get properties neither has on its own. Materials may be combined on a **bulk scale** — e.g. **reinforced concrete**: concrete is strong in compression, steel rods are strong in tension, so beams resist both — or **nanoparticles** may be embedded in a matrix (e.g. carbon nanotubes in sports equipment, C4.4). Fibreglass (glass fibres in polymer resin) is light, strong and can be moulded into boat hulls.

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Alloys

An **alloy** is a mixture of a metal with other elements (usually other metals, or carbon). Alloys have different properties from pure metals because atoms of **different sizes disrupt the regular layers** of the metal lattice, making it harder for layers to slide — so alloys are usually **harder and stronger** than the pure metal.

Pure Metal versus Alloy (model)



2D model: atoms drawn as hard spheres and delocalised electrons not shown

Alloy	Composition	Properties	Uses
Low-carbon (mild) steel	Iron + about 0.1–0.3% carbon	Strong, malleable, easily shaped; rusts	Car bodies, girders, wire
High-carbon steel	Iron + about 0.6–1.5% carbon	Very hard and strong but more brittle	Cutting tools, drill bits, springs
Stainless steel	Iron + chromium (~18%) + nickel (~8%)	Hard, resists corrosion	Cutlery, sinks, surgical instruments
Brass	Copper + zinc	Harder than copper; gold-coloured; does not corrode easily	Musical instruments, door handles, plumbing fittings

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Bronze	Copper + tin	Hard, tough, resists corrosion	Statues, medals, ship propellers
Aluminium alloys (e.g. duralumin)	Aluminium + copper, magnesium	Low density but much stronger than pure aluminium	Aircraft bodies, bike frames
Gold alloys	Gold + copper/silver (18 carat = 75% gold)	Harder than pure (24 carat) gold	Jewellery
Solder	Mainly tin (+ copper, silver)	Low melting point	Joining electrical components

By changing the composition, chemists can match an alloy to the specification of a new product.

> **Exam Tip: Ideas about Science (IaS3).** The alloy model explains why alloys are harder, but it is a simplification: real lattices are 3D, atoms are not hard spheres, and the model does not show the delocalised electrons or predict exactly how much harder an alloy will be.

Practical work: test a range of materials (e.g. for hardness by scratching, flexibility by bending, conductivity with a simple circuit, density by mass ÷ volume) and use the results to classify them.

C4.2 What Are the Different Types of Polymers?

> **Polymers** are very long-chain molecules made by joining many small molecules called **monomers**. They occur naturally and can be made synthetically. The **repeating unit** is the section of chain that repeats over and over.

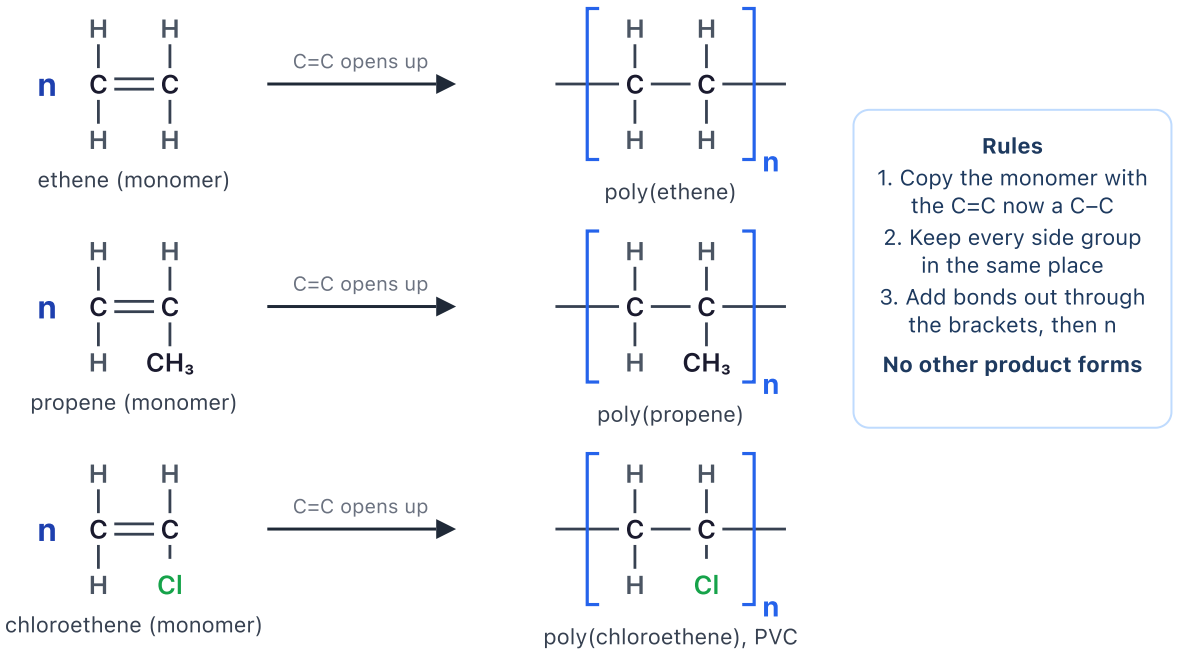
Addition Polymerisation

Monomers based on **alkenes** (made by cracking, C3.4) have a **C=C double bond** as their functional group. In addition polymerisation the double bonds **open up** and the monomers join

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together into a long chain. The polymer is the **only product**. These polymers are generally known as ‘plastics’.

Addition Polymerisation: Monomer → Polymer



Worked Example: A polymer has the repeating unit –[CH_2–CHCN]– (a CN group on the second carbon). Deduce the monomer.
Remove the brackets and the bonds sticking out, and put back the double bond between the two carbons: **CH_2=CHCN** (the monomer for an acrylic fibre).

Practical work: crack poly(ethene) by heating it over a hot catalyst and test the gas for unsaturation with bromine water (orange → colourless shows an alkene).

Condensation Polymerisation (HT)

(HT) Condensation polymers were developed to make substitutes for natural fibres such as wool and silk (e.g. polyesters and nylon).

- (HT) They usually form from **two different monomers**, each with **two functional groups** (one at each end) — that is the **minimum** needed, so the chain can keep growing in both directions.
- (HT) When two monomers join, the **–OH from a carboxylic acid group** and an **H atom from the other monomer** combine to form a **small molecule, water**, which is released at the same

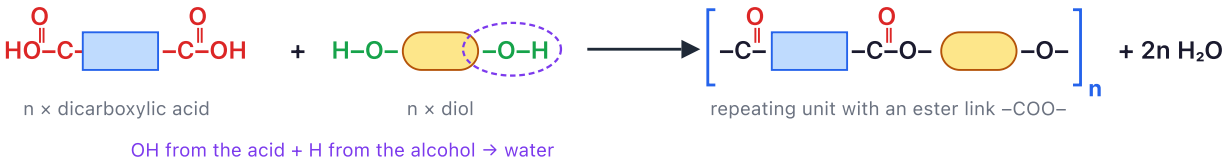
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time as each link forms.

- (HT) Dicarboxylic acid + **diol** (two –OH groups) → **polyester** (ester links, –COO–).
- (HT) Dicarboxylic acid + **diamine** (two –NH_2 groups) → **polyamide**, e.g. nylon (amide links, –CONH–).

(HT) Condensation Polymerisation Using Block Diagrams

Polyester



Polyamide (e.g. nylon)



Blocks stand for the carbon chains in the middle of each monomer; you do not need to recall the monomers' formulae.

Worked Example: (HT) Monomer A is HOOC–□–COOH and monomer B is HO–○–OH. Give the repeating unit and the other product.
Join the C of each acid group to the O of an alcohol group, removing OH from the acid and H from the alcohol each time.
Repeating unit: **–[OC–□–COO–○–O]–**; product: **water**, 2 molecules for each repeating unit formed (2n H_2O). Working backwards, split every ester link and add back –OH to the C=O side and –H to the O side.

(HT) Compare	Addition polymerisation	Condensation polymerisation
Monomers	One type, containing C=C	Usually two types, each with two functional groups
Products	Polymer only	Polymer + small molecule (water)

Repeating unit	Same atoms as the monomer	Atoms of both monomers minus water

Practical work: test the properties (strength, stretchiness, absorbency, effect of heat) of different polymer fibres, e.g. nylon, polyester, wool, cotton.

Naturally Occurring Polymers

- **DNA** is a polymer made from **four different monomers** called **nucleotides** (each with a different base: A, T, C, G). Genes are sections of DNA.
- **Proteins** are polymers of **amino acids** — similar in structure to polyamides (amide links join the amino acids).
- **Carbohydrates** such as **starch** and **cellulose** are polymers of **sugars** (glucose).

Practical work: break down starch using the enzyme amylase; use iodine solution (orange-brown → blue-black with starch) and Benedict's reagent (blue → brick-red precipitate on heating with sugars) to show starch disappearing and sugar forming.

C4.3 How Do Bonding and Structure Affect Properties of Materials?

The **bulk properties** of a material (strength, melting point, electrical and thermal conductivity, brittleness, flexibility, hardness, ease of reshaping) depend on the **types of bond** it contains, their **strength compared with intermolecular forces** and the **way the bonds are arranged**. A single atom does not have these properties: one copper atom is not 'shiny' or 'malleable' — the properties come from how vast numbers of atoms are bonded together. That is why the same atoms can make very different materials (diamond and graphite are both pure carbon).

Carbon: Chains, Rings and Families

- A carbon atom has 4 outer electrons, so it forms **four covalent bonds**.
- Carbon atoms can bond strongly to each other, forming **chains** (straight or branched) and **rings**, with single or double bonds, and can bond to H, O, N, Cl and others.

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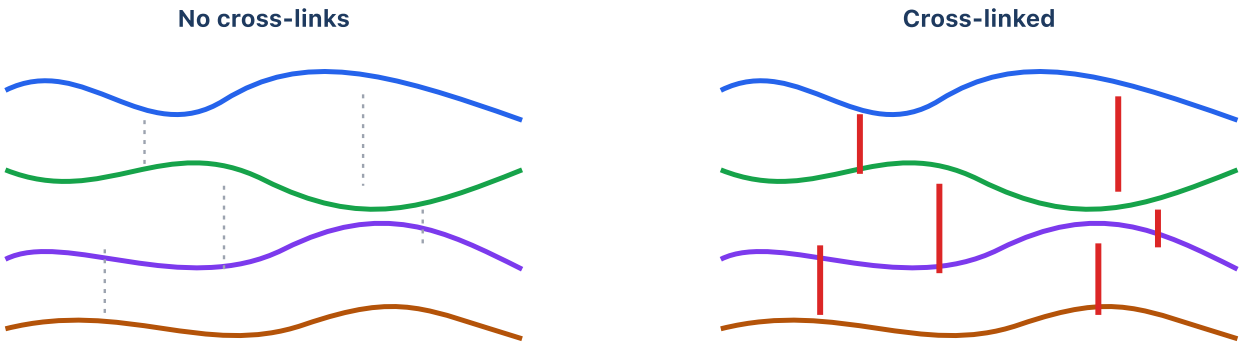
- This gives a **vast array** of natural and synthetic organic compounds, which fall into **families of similar compounds** — homologous series such as the alkanes (C3.4).

Polymers: Bonding and Properties

Polymer molecules have the same strong covalent bonds as simple molecules, but because the chains are so long there are **many more intermolecular forces** between neighbouring chains. The strength of these forces decides the properties of the solid:

- **Longer chains** → more intermolecular forces and more tangling → stronger, higher melting (softening) point.
- **Unbranched chains** pack closely → stronger forces → denser, stiffer, harder (HDPE). **Branched chains** cannot pack closely → weaker, more flexible, lower density (LDPE).
- **Cross-links** (covalent bonds between chains) stop the chains sliding → rigid, hard, do not soften on heating (they char instead) — e.g. vulcanised rubber, melamine.
- **Plasticisers** are small molecules that push chains apart, weakening the forces → more flexible (e.g. soft PVC).

Polymer Chains with and without Cross-Links



weak intermolecular forces (dotted) between chains
chains slide when heated → soften, can be reshaped

strong covalent cross-links (red) hold chains in place
rigid and hard; do not soften on heating

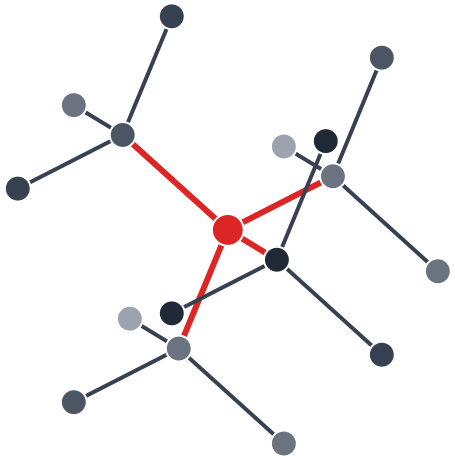
Exam Tip: Ideas about Science (IaS2). Look for patterns in data about polymers. For example (illustrative data): poly(ethene) samples with average chain lengths of 1000, 5000 and 20 000 units softening at 95 °C, 110 °C and 125 °C show that longer chains give higher softening temperatures — explained by more intermolecular forces between longer chains.

Giant Covalent Structures

In a **giant covalent structure** a very large number of atoms are joined by covalent bonds in a continuous three-dimensional network — there are no separate molecules. To melt one, strong covalent bonds must be broken, so melting points are **very high**. Carbon forms several giant structures called **allotropes** (different structural forms of the same element), including diamond and graphite. Silicon dioxide (sand) is another giant covalent structure.

Giant Covalent Structures of Carbon: Diamond and Graphite

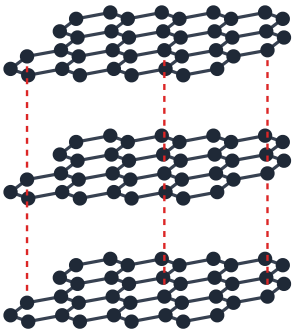
Diamond



each C atom bonded to **4** others (tetrahedral)
rigid 3D network of strong covalent bonds
no delocalised electrons → does not conduct

(red atom shown with its 4 bonds; network continues in all directions)

Graphite



each C atom bonded to **3** others in hexagonal layers
one delocalised electron per C atom → conducts
layers slide over each other → soft, slippery

Property	Diamond	Graphite
Melting point	Very high (above 3500 °C) — many strong covalent bonds to break	Very high — strong covalent bonds within the layers must break
Hardness	Hardest natural substance: rigid 3D network, each C held by 4 bonds → cutting tools, drill tips	Soft: weak forces between layers let them slide → pencil 'lead'
Electrical conductivity	Does not conduct: all 4 outer electrons are in bonds	Conducts: each C uses 3 electrons in bonds; the 4th is delocalised and moves along the layers → electrodes
Lubricating action	—	Layers slide easily over each other → dry lubricant

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The diagrams above show a 3D structure on a flat page. You should be able to go from a 3D model to a 2D drawing and back (M5b) — e.g. spot that each diamond atom has 4 neighbours even though some bonds look shorter because they point towards or away from you.

Comparing All the Types of Structure

Structure	Bonding and arrangement	Melting point	Conducts?	Other properties
Ionic (NaCl)	Giant lattice of oppositely charged ions; strong electrostatic attraction in all directions (C2.3)	High	Only when molten or dissolved (ions free to move)	Hard but brittle
Simple molecular (CH_4, H_2O)	Strong covalent bonds within small molecules; weak intermolecular forces between them	Low	No	Often gases or liquids
Giant covalent (diamond, SiO_2)	Huge 3D network of strong covalent bonds	Very high	No (graphite and graphene are exceptions)	Hard, insoluble
Polymer (poly(ethene))	Long covalent chains; intermolecular forces (or cross-links) between chains	Low to moderate; soften over a range	No	Flexible or rigid depending on chains
Metallic (copper)	Giant lattice of positive ions in a sea of delocalised electrons (C3.1)	Usually high	Yes, as solid and liquid	Malleable, ductile

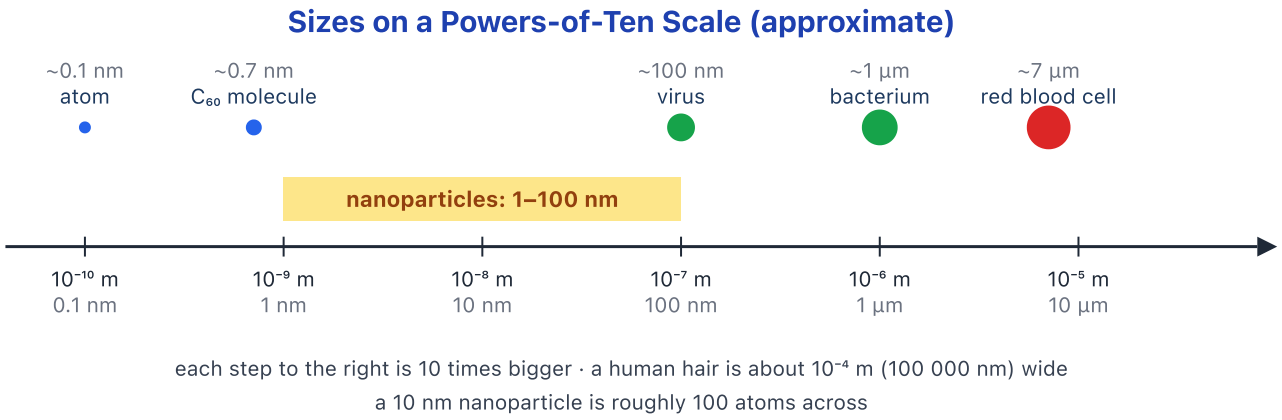
Practical work: test the properties (melting, solubility, conductivity as solid and in solution) of simple covalent compounds, giant ionic and giant covalent substances, metals and polymers.

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Exam Tip: Ideas about Science (IaS3). Structure models (ball and stick, layer diagrams) help chemists predict properties of new materials, but they have limits: bonds are not sticks, atoms are not spheres of the sizes shown, and 2D pictures hide parts of 3D structures.

C4.4 Why Are Nanoparticles So Useful?

Nanotechnology is the use and control of structures that are very small: **1 to 100 nanometres** in size. 1 nm = 1×10^{-9} m (a billionth of a metre). **Nanoparticles** have a similar scale to individual molecules.



each step to the right is 10 times bigger · a human hair is about 10^{-4} m (100 000 nm) wide
a 10 nm nanoparticle is roughly 100 atoms across

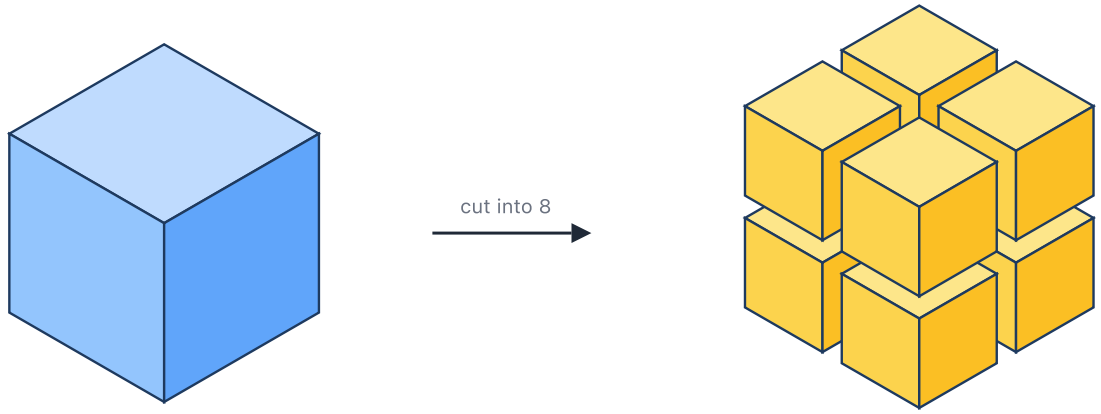
Worked Example: Standard form and ratio. An atom is about 1×10^{-10} m across and a nanoparticle is 2×10^{-8} m across. How many atoms fit across the nanoparticle?
Ratio = $(2 \times 10^{-8}) \div (1 \times 10^{-10}) = 2 \times 10^{2}$ = **200 atoms**. In nanometres: 20 nm ÷ 0.1 nm = 200.
Order these from smallest: 5×10^{-7} m, 3×10^{-9} m, 8×10^{-8} m → 3×10^{-9} m (3 nm) < 8×10^{-8} m (80 nm) < 5×10^{-7} m (500 nm). Only the first two are in the nano range.

Surface Area to Volume Ratio

For a cube of side L: **surface area = $6L^2$**, **volume = L^3**, so **SA : V = 6/L**. As particles get smaller, the surface area to volume ratio gets **bigger**: making the side 10 times smaller makes SA : V 10 times larger.

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Cutting a Cube Up: Same Volume, More Surface



one cube, side 2 cm
SA = 6 × 2² = 24 cm² · V = 2³ = 8 cm³
SA : V = 24 : 8 = 3 : 1

eight cubes, side 1 cm
SA = 8 × 6 × 1² = 48 cm² · V = 8 × 1³ = 8 cm³
SA : V = 48 : 8 = 6 : 1

Worked Example: Find SA : V for a cubic nanoparticle of side 10 nm and compare it with a cube of side 1 µm.
10 nm cube: SA = 6 × (10)² = 600 nm²; V = 10³ = 1000 nm³; SA : V = 0.6 nm^{-1} (= 6 × 10⁸ m^{-1}).
1 µm = 1000 nm cube: SA : V = 6 ÷ 1000 = 0.006 nm^{-1}.
The nanoparticle's ratio is **100 times** larger (side is 100 times smaller).

Because so many of their atoms are at the surface, nanoparticles behave differently from the same material in bulk:

- They are **excellent catalysts** — reactions happen at surfaces, so less material does the same job.
- They can be **more reactive** than bulk material, and can have new properties (e.g. nanoparticles of titanium dioxide or zinc oxide in sunscreen block UV but look transparent; silver nanoparticles are antibacterial).
- Their tiny size lets them **penetrate biological tissue** and be built into other materials to modify their properties.

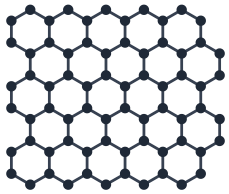
Fullerenes and Graphene

Fullerenes are molecules of carbon atoms arranged in hollow **balls or tubes**, made of rings of carbon atoms (mainly hexagons, with pentagons where the surface curves). Each carbon is bonded to three others.

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Graphene, a Carbon Nanotube and Buckminsterfullerene (C_{60})

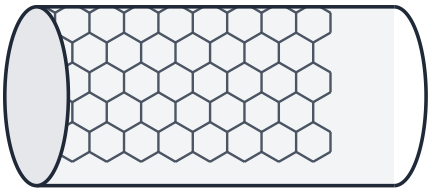
Graphene



a single layer of graphite — one atom thick
each C bonded to 3 others
very strong; conducts electricity and heat well

Carbon nanotube

a rolled-up sheet of graphene (hollow)



cylinder of hexagons, ~1 nm wide
but can be very long
very high tensile strength; conducts; reinforces composites (e.g. sports equipment)

Buckminsterfullerene C_{60}



hollow ball: 60 C atoms in 12 pentagons + 20 hexagons
(front face shown; red = a pentagon)
simple molecule ~1 nm across; can carry drug molecules

Material	Structure	Properties → uses
Buckminsterfullerene, C_{60}	Hollow ball of 60 C atoms (simple molecule)	Cage can carry small molecules → delivering drugs into the body; slippery molecules → lubricants; catalysts
Carbon nanotubes	Cylinders of carbon hexagons, nanometres wide but long	Very strong, light, conduct electricity → reinforcing composites (tennis rackets, bike frames), electronics; tiny pores → **molecular sieves**
Graphene	Single layer of graphite, one atom thick	Very strong, transparent, high electrical and thermal conductivity → flexible electronics, touchscreens, batteries, composites

Exam Tip: Ideas about Science (IaS3). Developing fullerenes and graphene relied on imaginative thinking. C_{60} was discovered in 1985 when Kroto, Curl and Smalley imagined a football-shaped cage to explain an unexpected result; graphene was first isolated in 2004 by Geim and Novoselov — famously using sticky tape to peel layers off graphite.

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Risks of Nanoparticles

- **(a) Possible health effects due to size and surface area:** nanoparticles can be breathed in, absorbed through skin or cross cell membranes; their large surface area may make them reactive or catalyse harmful reactions in the body; some long, thin nanotubes may behave like asbestos fibres in the lungs.
- **(b) Why there is more data about uses than about health effects:** nanotechnology is new; companies fund research into useful (profitable) products; health effects may take many years to show; tiny particles are hard to detect and track; each type, size and shape can behave differently; testing on humans raises ethical problems.
- **(c) Relative risks and benefits:** nanoparticles in sunscreen give a large benefit (protection from skin cancer) for a probably small risk, so many people accept them; nanosilver in socks to reduce odour offers a small benefit while silver washed into rivers could harm useful microorganisms.

> **Exam Tip: Ideas about Science (IaS4).** Because not much is known about some nanoparticles' effects on the body, judgements about each use depend on balancing the perceived benefit against the risk. In a 6-mark answer, discuss a benefit, a risk, the lack of data, and who bears the risk, then give a justified conclusion.

C4.5 What Happens to Products at the End of Their Useful Life?

Corrosion and Rusting

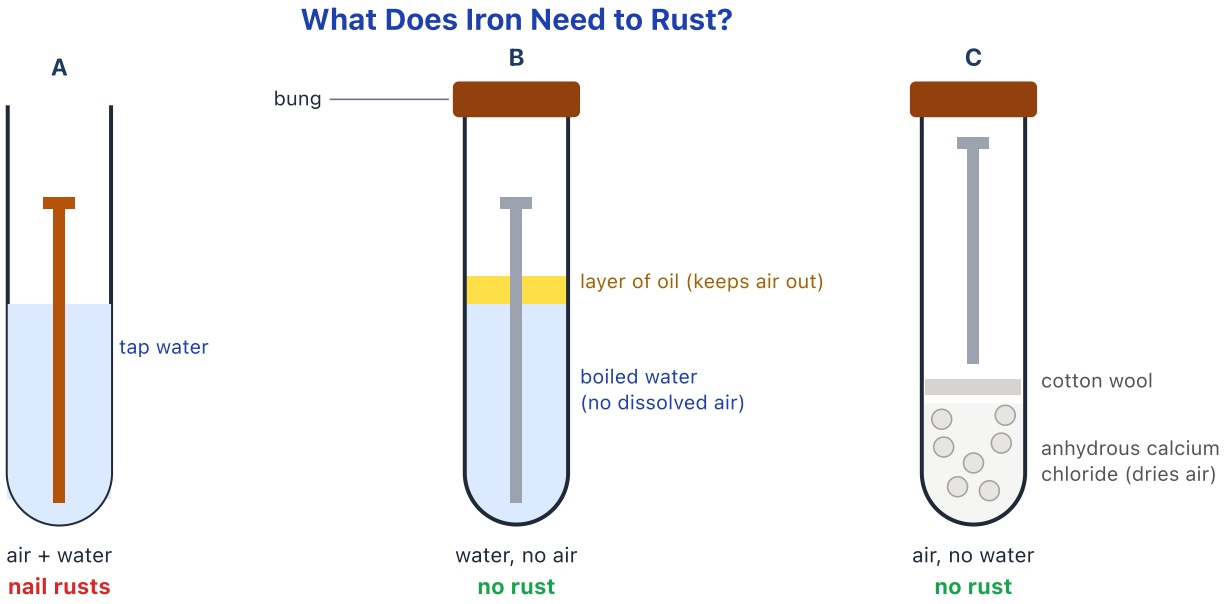
Iron is the most widely used metal in the world, but iron products have a limited life because iron **corrodes**. **Corrosion** is the gradual reaction of a metal with substances in its surroundings. Rusting is the corrosion of iron and steel; it is an **oxidation** reaction that needs **both oxygen (air) and water**.

$$\text{iron} + \text{oxygen} + \text{water} \rightarrow \text{hydrated iron(III) oxide (rust)}$$

Salt water speeds rusting up. Rust is flaky and crumbles away, exposing fresh iron underneath, so rusting continues until the iron is eaten through. (Aluminium also corrodes, but forms a thin, tough

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oxide layer that protects the metal beneath.)



Practical work: investigate the factors needed for rusting (tubes as above, left for about a week) or for the corrosion of other metals, and the effectiveness of corrosion prevention (e.g. nails painted, greased, wrapped in zinc or magnesium, wrapped in copper, left in salt water).

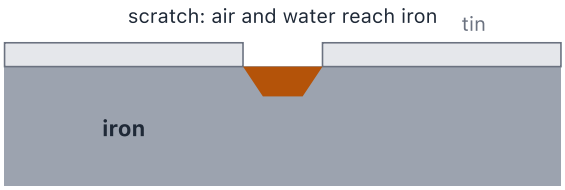
Preventing Corrosion

- **Barrier methods** keep oxygen and water away from the metal: **paint** (bridges, cars), **oil or grease** (moving parts, bike chains), **plastic coating** (garden furniture, wire), **electroplating** with a less easily corroded metal (tin cans: steel coated with tin). If the barrier is scratched, the iron underneath starts to rust.
- **Sacrificial protection:** iron is attached to, or coated with, a **more reactive metal** such as **zinc** or **magnesium**. The more reactive metal oxidises **in preference to iron**, so the iron is protected **even if the coating is damaged**. Examples: **galvanising** (zinc coating on steel buckets, fences, car bodies), zinc or magnesium blocks bolted to ships' hulls and underground pipes (replaced as they corrode away).

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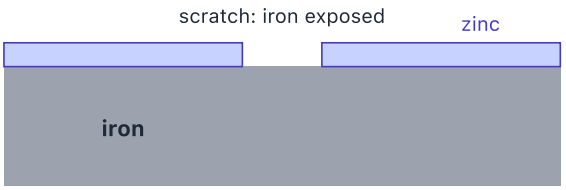
When the Coating Is Scratched

Tin plating (barrier only)



iron rusts at the scratch
(tin is less reactive than iron, so cannot protect it)

Galvanising (zinc: sacrificial)



zinc corrodes instead; iron protected
(zinc is more reactive, so it is oxidised first)

> **Exam Tip: Ideas about Science (IaS4).** Preventing corrosion is a scientific solution that improves **sustainability**: products last longer, so fewer metal ores are mined and less energy is used making replacements — a positive outcome in a life cycle assessment.

Oxidation and Reduction in Terms of Oxygen

> **Oxidation** is the **gain of oxygen**; **reduction** is the **loss of oxygen**. They always happen together (redox).

- Rusting: iron gains oxygen → iron is **oxidised**.
- $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$: iron(III) oxide loses oxygen → **reduced**; carbon monoxide gains oxygen → **oxidised**.
- $2Mg + O_2 \rightarrow 2MgO$: magnesium is oxidised.

Oxidation and Reduction in Terms of Electrons (HT)

> (HT) **Oxidation is loss of electrons; reduction is gain of electrons** (OIL RIG).

- (HT) In rusting, iron atoms lose electrons: $Fe \rightarrow Fe^{3+} + 3e^-$ (iron is **oxidised**); oxygen gains electrons: $O_2 + 4e^- \rightarrow 2O^{2-}$ (oxygen is **reduced**).
- (HT) In sacrificial protection, zinc is oxidised instead of iron: $Zn \rightarrow Zn^{2+} + 2e^-$.

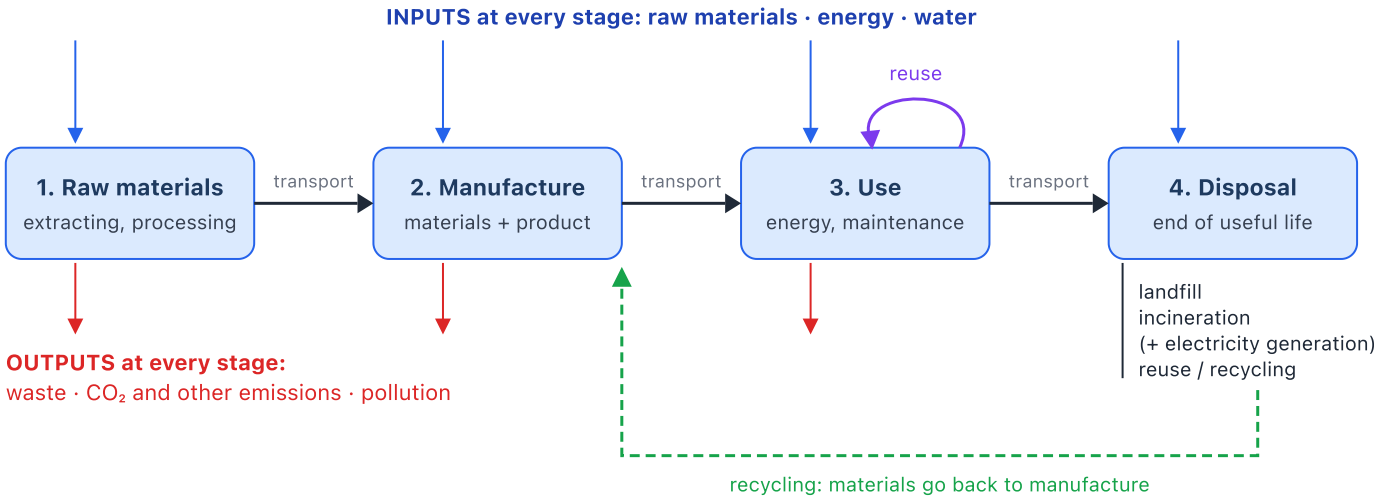
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- (HT) In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$: zinc loses electrons (oxidised); Cu^{2+} ions gain electrons (reduced).

Life Cycle Assessment (LCA)

> A **life cycle assessment** considers the overall impact of making, using and disposing of a product: the use of **resources (raw materials, water, energy)** and the **environmental impact** at every stage, from raw materials to disposal, including **transport** between stages.

Stages of a Life Cycle Assessment



Disposal Options

Option	Advantages	Disadvantages
Landfill	Cheap and simple	Uses land; non-biodegradable materials last for centuries; biodegradable waste rots to release methane (a greenhouse gas); toxic substances can leach into groundwater
Incineration	Reduces volume of waste a lot	Releases CO_2 and possibly toxic gases (needs cleaning); leaves ash; the materials are lost

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Incineration with electricity generation (energy from waste)	Energy released is used to generate electricity, saving some fossil fuel	Still releases CO_2; materials lost; plants are expensive to build
Reuse	Uses the least energy (only cleaning/repair)	Product must be durable; collection and cleaning needed
Recycling	Conserves raw materials; usually less energy than making from new	Collection, sorting and cleaning cost energy and money; quality may fall

Biodegradable materials (paper, wood, cotton, some bioplastics) are broken down by microorganisms; **non-biodegradable** materials (most addition polymers such as poly(ethene), glass) are not, so they persist in landfill and the environment.

Why LCAs are not fully secure (IaS4): there is not always enough reliable data; some effects (e.g. visual pollution) need value judgements; and people do not always follow recommended disposal advice, so the real end-of-life route differs from the assumed one. LCAs can also be selectively written to support a company's claims.

Interpreting LCA Data

Illustrative data (per 1000 drinks containers)	Glass bottle	PET bottle	Aluminium can
Energy to make / MJ	2500	1200	1600
Water used / dm^3	4000	2500	3500
CO_2 from transport / kg	60	15	12
Can be reused?	Yes (refillable)	Limited	No
Recycling rate assumed	70%	60%	75%

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Worked Example: (Using the illustrative data above.) Which container has the lowest impact?
PET uses the least energy and water to make; the aluminium can has the lowest transport emissions (lightest per drink) and is widely recycled; glass is heaviest (high transport CO_2) and needs most energy, *but* can be refilled many times — if each bottle is reused 10 times its manufacturing energy per use falls to about 250 MJ per 1000 drinks, the lowest of all.
Conclusion: the 'best' choice depends on assumptions about reuse, recycling rates and transport distances — always state them.

Reusing and Recycling PET Drinks Bottles

PET (poly(ethylene terephthalate)) is a polyester. Bottles can be **reused** (refilled after cleaning) or **recycled**:

1. Collected (kerbside or bottle banks) and transported to a plant.
2. Sorted from other plastics (by recycling code 1, or automatic scanners) and often by colour.
3. Caps and labels removed; bottles washed to remove impurities.
4. Shredded into flakes; caps (a different polymer) separated by flotation — PET is denser than water and sinks.
5. Melted and made into pellets, then into new products: fleece clothing fibres, carpets, filling for sleeping bags, strapping, and new food-grade bottles.

Why PET recycling is viable: PET bottles are used in huge numbers, so there is a large, steady supply; they are easy to identify and separate; PET is thermosoftening, so it can be melted and reshaped; there is strong demand for products made from it; and it saves crude oil and energy compared with making new PET.

Factors in Recycling Decisions

Recycling conserves resources such as crude oil and metal ores, but will not on its own meet future demand unless habits change. Whether recycling a material is worthwhile depends on:

- the **finite nature** of the raw material deposits (metal ores, crude oil);
- the **availability** of the used material to collect;
- the economic and practical costs of **collection and sorting**;

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- **removing impurities** (mixed plastics, alloys, food contamination);
- **energy used** in transport and processing;
- the **scale of demand** for the recycled product;
- the **environmental impact** of the recycling process itself.

Products from metal ores	Products from crude oil (plastics)
Metals can be melted and reused again and again without losing quality. Recycling aluminium needs only about 5% of the energy needed to extract it from its ore by electrolysis, so it is very worthwhile. Steel is easily separated with magnets.	Many different polymers are mixed together and are hard to sort; contamination is common; chains can shorten on reprocessing so quality falls ('downcycling'); some cannot be remelted (thermosets); low price of new plastic reduces demand for recycled plastic.

Exam Tip: Ideas about Science (IaS4). Products made from recycled materials do *not* always have a lower environmental impact than those made from new resources — e.g. if collection involves long transport distances or the cleaning uses lots of energy and water. Use LCAs to compare the sustainability of products and processes.

C4 Summary & Key Equations

- Choose materials by matching measured properties (strength, stiffness, density, melting point, conductivity…) to a product's specification
- Composites combine properties; alloys are harder because different-sized atoms stop layers sliding
- Addition polymers: C=C opens, polymer only; condensation polymers (HT): two functional groups per monomer, water also formed
- DNA = 4 nucleotides; proteins = amino acids; starch/cellulose = sugars
- Properties come from bonding and structure, not from individual atoms
- Diamond: 4 bonds each, hard, non-conductor; graphite: 3 bonds, layers, conducts, lubricant
- Nano = 1–100 nm; SA : V = 6/L for a cube; smaller particles → larger SA : V

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- Rusting needs oxygen + water; barrier vs sacrificial (more reactive Zn/Mg) protection
- Oxidation = gain of O (loss of e^-, HT); reduction = loss of O (gain of e^-, HT)
- LCA: raw materials → manufacture → use → disposal, with transport, energy and water at each stage

> **Exam Tip:** In "choose a material" questions, quote numbers from the data and link each property to the job it must do — "low density (0.92 g/cm³) so the bottle is light to transport" scores; "it is light" does not.

Specification Checklist

Ref	I can...	Tier
C4.1.1	compare physical properties of glass, clay ceramics, polymers, composites and metals quantitatively	F + H
C4.1.2	relate properties to uses and select materials	F + H
C4.1.3	describe the composition, properties and uses of important alloys including steel	F + H
C4.2.1	recall principles of addition polymerisation (functional group, repeating unit)	F + H
C4.2.2	deduce an addition polymer from its monomer and vice versa	F + H
C4.2.3	explain condensation polymerisation	(HT)
C4.2.4	recall DNA (four nucleotides) and polymers based on sugars and amino acids	F + H
C4.3.1	explain bulk properties from bonds, bond strengths versus intermolecular forces and arrangement	F + H

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C4.3.2–3	recall carbon forms four bonds; explain the variety of organic compounds	F + H
C4.3.4	describe bonding in polymers and link it to their properties	F + H
C4.3.5–6	describe giant covalent structures; explain properties of diamond and graphite	F + H
C4.3.7	represent 3D structures in 2D and vice versa (M5b)	F + H
C4.3.8	compare ionic, simple molecular, giant covalent, polymer and metallic bonding	F + H
C4.4.1–3	compare nano dimensions with atoms; SA : V; link nanoparticle properties to uses	F + H
C4.4.4	explain properties of fullerenes and graphene from structure	F + H
C4.4.5	explain risks of nanoparticles (a)–(c)	F + H
C4.4.6–9	estimate sizes in nm; standard form; ratios; SA and volume of cubes (M1d, M1b, M1c, M5c)	F + H
C4.5.1	describe corrosion and its prevention by barriers and sacrificial protection	F + H
C4.5.2	explain oxidation and reduction as gain or loss of oxygen	F + H
C4.5.3	explain oxidation and reduction as loss or gain of electrons	(HT)
C4.5.4–5	describe the principles of LCA (a)–(c) and interpret LCA data	F + H

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C4.5.6	describe reuse and recycling of PET bottles and why it is viable	F + H
C4.5.7	evaluate recycling decisions for crude-oil and metal-ore products	F + H

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